

Answer Key — Physics Grade 10

Sub-Strand 1.2: Pressure • Lesson 2: What Is Pressure? Measuring the Invisible Push

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5
B	C	B	C	C

USE AFTER: EXPLAIN PHASE — Formal definition $P = F/A$, after the book vs pencil demonstration

Q1. A book and a pencil are pressed onto your palm with the same force. Why does the pencil tip hurt more?

- A. The pencil is heavier than the book
- B. The same force acts on a much smaller area, so the pressure is greater**
- C. The pencil pushes with a bigger force
- D. The book spreads the force over a smaller area

Correct answer: B. $P = F/A$. With F the same, a smaller contact area gives a larger pressure. The masses and forces were deliberately kept equal.

USE AFTER: OBSERVE PHASE — Part B: shoe-pressure investigation (cm^2 to m^2 conversion)

Q2. A shoe sole covers 150 cm^2 . What is this area in square metres?

- A. 1.5 m^2
- B. 0.15 m^2
- C. 0.015 m^2**
- D. 15 m^2

Correct answer: C. $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, so $150 \div 10\,000 = 0.015 \text{ m}^2$. This conversion is the error the lesson plan flags as most common.

USE AFTER: EXPLAIN PHASE — Problem 1: maize bag on a pallet (Eldoret warehouse)

Q3. A maize bag on a pallet pushes 225 N onto each leg. Each leg touches the floor over 0.0025 m^2 , so the pressure is $90\,000$ _____. Which unit completes the answer?

- A. Newtons (N)
- B. Pascals (Pa), which are N/m^2**
- C. Kilograms (kg)
- D. Square metres (m^2)

Correct answer: B. Pressure = force $\div$ area, so its unit is N/m^2 , called the pascal (Pa). Newton is force, kg is mass, m^2 is area.

USE AFTER: EXPLAIN PHASE — Problem 2: rungu vs crutch

Q4. A rungu tip (2 cm^2) and a crutch base (50 cm^2) each press down with 150 N . Which statement is correct?

- A. They exert the same pressure because the force is the same
- B. The crutch exerts higher pressure because it is larger
- C. The rungu exerts higher pressure because its area is smaller**
- D. Neither exerts pressure because they are not moving

Correct answer: C. Rungu: $150 \div 0.0002 = 750\,000 \text{ Pa}$. Crutch: $150 \div 0.005 = 30\,000 \text{ Pa}$. Same force, smaller area, much higher pressure. So the crutch is safer on a soft floor.

Q5. In which directions does atmospheric pressure push on the jerrycan?

- A. Only downward, on the top
- B. Only on the sides
- C. Inward on every surface: top, bottom and all sides**
- D. Outward from the inside only

Correct answer: C. Atmospheric pressure acts on every exposed surface, from all directions. The lesson plan flags students who draw arrows only on top as holding a key misconception.